1. Discrete Distributions

	Bern(p)	Binom(n,p)	Geom(p)	Pois(λ)	Disc Unif $\{1, \dots, n\}$
Range	{0,1}	{0,1,...,n}	{1,2,3,...}	{0,1,2,...}	{1,...,n}
PMF	$p^x(1-p)^{1-x}$	$\binom{n}{k}p^k(1-p)^{n-k}$	$(1-p)^{k-1}p$	$e^{-\lambda}\frac{\lambda^k}{k!}$	$\frac{1}{n}$
E[X]	p	np	1/p	λ	$\frac{n+1}{2}$
Var	p(1−p)	np(1−p)	(1−p)/p²	λ	$\frac{n^2-1}{12}$
MGF	$(1-p)+pe^t$	$[(1-p)+pe^t]^n$	$\frac{pe^t}{1-(1-p)e^t}$	$e^{\lambda(e^t-1)}$	$\frac{1}{n}\sum_{k=1}^{n} e^{tk}$

Binom = sum of n iid Bern(p) | Indicator: $I_A \sim$ Bern($P(A)$) | Geom: memoryless (discrete) | $E[X] = \sum x \cdot P(X = x)$ | LOTUS: $E[g(X)] = \sum g(x)P(X = x)$ MGF domains: Geom $t < -\ln(1-p)$ | Bern/Binom/Pois/Disc Unif: all t | General disc unif on $\{x_1, \dots, x_n\}$: $E[X] = \frac{1}{n}\sum x_i$

2. Variance & Linearity

$E[\sum a_i X_i] = \sum a_i E[X_i]$ | $E[aX + b] = aE[X] + b$ $\text{Var}(X) = E[X^2] - (E[X])^2$ | $E[X^2] = \text{Var}(X) + (E[X])^2$ $\text{Var}(aX + b) = a^2\text{Var}(X)$ $\text{Var}(\sum a_i X_i) = \sum a_i^2 \text{Var}(X_i)$ **only when independent** | E is always linear, Var is not

3. Continuous RVs

$f(x) \geq 0$, $\int f(x)dx = 1$ (can be more than 1) | $P(X = a) = 0$ $F(x) = \int_{-\infty}^{x} f(t)dt = P(X \leq x)$ | $f(x) = F'(x)$ | $P(a < X < b) = F(b) - F(a)$ | endpoints don't matter | Normalizing: $f(x) = c \cdot h(x)$, set $\int c \cdot h(x)dx = 1$ | LOTUS: $E[g(X)] = \int g(x)f(x)dx$ Quantiles: $P(X \leq q_\alpha) = \alpha$ | if invertible, $q_\alpha = F^{-1}(\alpha)$ | Median = $q_{1/2}$ | IQR = $q_{3/4} - q_{1/4}$ Nonneg RVs: $E[X] = \int_0^\infty P(X > x)dx = \int_0^\infty [1 - F(x)]dx$ (other E[X] way)

4. Continuous Distributions

Uniform(a, b): Range $[a, b]$ | $f = \frac{1}{b-a}$ | $F = \frac{x-a}{b-a}$ | $E = \frac{a+b}{2}$ | Var $= \frac{(b-a)^2}{12}$

Normal(μ, σ^2): Range $\mathbb{R}$ | $f = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$ | $E = \mu$ | Var $= \sigma^2$ $Z = (X - \mu)/\sigma \sim N(0, 1)$ | $P(a < X < b) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$ | $c = \mu + \sigma\Phi^{-1}(\alpha)$ $aX + b \sim N(a\mu + b, a^2\sigma^2)$ (linear transform stays Normal) | 68-95-99.7 rule

Exp(β): Range $[0, \infty)$ | $f = \beta e^{-\beta x}$ | $F = 1 - e^{-\beta x}$ | $P(X > x) = e^{-\beta x}$ | $E = 1/\beta$ | Var $= 1/\beta^2$ $q_\alpha = -\frac{1}{\beta}\ln(1 - \alpha)$ | Memoryless: $P(X > t + s | X > t) = P(X > s)$ | Exp = Gamma($1, \beta$)

Gamma(α, β): Range $(0, \infty)$ | $f = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ | $E = \alpha/\beta$ | Var $= \alpha/\beta^2$ | α = shape, β = rate

Beta(a, b): Range $(0, 1)$ | $f = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$ | **Gamma fn:** $\Gamma(a+1) = a\Gamma(a)$ | $\Gamma(n) = (n-1)!$ | $\Gamma(1/2) = \sqrt{\pi}$

5. Kernel recognition

$\int_0^\infty x^{\alpha-1}e^{-\beta x}dx = \frac{\Gamma(\alpha)}{\beta^\alpha}$ (gamma kernel: power = $\alpha - 1$, rate = β) $\int_{-\infty}^{\infty} \exp(-\frac{(x-\mu)^2}{2\sigma^2})dx = \sqrt{2\pi\sigma^2}$ (Gaussian kernel: complete the square) Complete square: $-ax^2 + bx = -a(x - \frac{b}{2a})^2 + \frac{b^2}{4a}$

6. MGFs

$M_X(t) = E[e^{tX}]$ | $M(0) = 1$ | $M'(0) = E[X]$ | $M''(0) = E[X^2]$ | Var $= M''(0) - (M'(0))^2$ Same MGF = same dist | $M_{aX+b}(t) = e^{bt}M_X(at)$ | Indep: $M_{X+Y} = M_X \cdot M_Y$ Domains: Exp $M(t) = \frac{\beta}{\beta-t}$, $t < \beta$ | Gamma $M(t) = (\frac{\beta}{\beta-t})^\alpha$, $t < \beta$ | Normal/Pois: all t

7. Transformations

CDF method: (1) Range(Y) (2) $F_Y(y) = P(g(X) \leq y)$ (3) Solve for X, use F_X (4) Differentiate Step 2: isolate X inside F_X: e.g. $P(2X+3 \leq y) = P(X \leq \frac{y-3}{2}) = F_X(\frac{y-3}{2})$. Then differentiate with chain rule. Increasing g: inequality same | Decreasing g: flips (e.g. $P(1/X \leq y) = P(X \geq 1/y) = 1 - F_X(1/y)$) **Monotone shortcut:** $f_Y = f_X(g^{-1}(y)) \cdot |(g^{-1})'(y)|$ Non-monotone ($Y = X^2$): use CDF method, $P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y})$, simplifies if $X > 0$

8. Joint Distributions

Discrete joint: same formulas, but **sums replace integrals** | $p_X(x) = \sum_y p_{XY}(x,y)$ | $p_{X|Y}(x|y) = p_{XY}(x,y)/p_Y(y)$| $f_X(x) = \int f_{XY}(x,y)dy$ (marginal) | $f_{X|Y}(x|y) = f_{XY}(x,y)/f_Y(y)$ (conditional)| $f_{XY} = f_{Y|X} \cdot f_X = f_{X|Y} \cdot f_Y$ (decomposition) | $f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)}$ (Bayes)| Independence: $f_{XY} = f_X \cdot f_Y$ for all $(x,y) \Leftrightarrow$ conditional = marginal| Hierarchical: given $X \sim P_X$, $Y|X = x \sim P_{Y|X}$, joint = $f_{Y|X}(y|x) \cdot f_X(x)$| $E[X|Y=y] = \int x f_{X|Y}(x|y)dx$ (function of y) | $E[g(X,Y)] = \iint g \cdot f_{XY} dx dy$| If indep: $E[g_1(X)g_2(Y)] = E[g_1(X)] \cdot E[g_2(Y)]$ | **SKETCH JOINT RANGE FIRST**| $0 < x < y < \infty$: fix $y \to x \in [0,y]$; fix $x \to y \in [x,\infty)$ | $0 < y < x < 1$: fix $x \to y \in [0,x]$; fix $y \to x \in [y,1]$

9. Sums/Averages of IID RVs

X_i iid, mean μ, var σ^2, mgf $M(t)$ | $S_n = \sum X_i$ | $\bar{X}_n = S_n/n$ S_n: $E = n\mu$ | Var $= n\sigma^2$ | MGF $= M(t)^n$ $\bar{X}_n$: $E = \mu$ | Var $= \sigma^2/n$ | MGF $= M(t/n)^n$ Bern $\to$ Binom(n,p) | Pois $\to$ Pois$(n\lambda)$ | Exp $\to$ Gamma(n,β) Gamma$(\alpha,\beta) \to$ Gamma$(n\alpha,\beta)$ | $N(\mu,\sigma^2) \to N(n\mu, n\sigma^2)$, avg $N(\mu, \sigma^2/n)$

10. Reference & Tricks

$n = 1$ raw = mean | $n = 2$ central = var | $n = 3$ std = skewness | $n = 4$ std = kurtosis From $E[X^n]$ Skew: 0 symm, > 0 right tail, < 0 left | $\sum r^k = 1/(1-r)$ | $e^x = \sum x^k/k!$ | $(x+y)^n = \sum \binom{n}{k} x^k y^{n-k}$ $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ | $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ | use for disc unif mgf $\to$ mean/variance Conditioning: $P(X = k|A) = \frac{P(X=k,\, A)}{P(A)}$ | **Key pattern:** counting RV = sum of indicators $\to$ use linearity for E, independence for Var **MGF exists = all moments finite** (not converse) | **same MGF = same dist** (same moments does NOT)